

Anthony Zhao

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I am a US Citizen

EDUCATION

University of Toronto

Toronto, ON

Bachelor of Science in Computer Science and Statistics

Expected Jun. 2028

- Relevant Courses: Machine Learning, Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Web Development, Data Science, Statistics, Agile

EXPERIENCE

Software Engineering Intern: ML Infra

May 2026 – August 2026

Peripheral Labs

Toronto, ON

- Unlocked **\$300K** in **GCP** credits by building training infrastructure from scratch: **Slurm** on Cluster Toolkit, a custom GCE image for **CUDA**, and **gcsfuse** storage with local SSD caching
- Enabled multi-node **H100/H200** training on **Slurm** at **~300+ GB/s** all-reduce bandwidth by configuring **TCPXO/RDMA** fabric
- Reduced GPU wait time by **80%** by automating **Slurm** job scheduling across regions on spot and flex capacity with custom availability dashboards, so researchers run multi-GPU training wherever capacity exists
- Built multi-view video data-loading infra streaming **S3** to **EC2**, cutting training data load time from **30 min** to **20s (90x)** and adopted as the sole access path for video data across **4** downstream ML pipelines
- Made the pipeline hardware-bound instead of software-bound by replacing single-stream downloads with **parallel ranged reads**, leaving **NVDEC** GPU decode as the only bottleneck at **80%+** GPU utilization
- Shipped a config switch to let decoded frames live in GPU memory or offload to system RAM, enabling **2x** larger batches with fully randomized sampling

Software Engineering Intern

Nov 2025 – Mar 2026

PhrasIQ

San Francisco, CA

- Built an **LLM**-based synthetic data generation pipeline **microservice** for training and fine-tuning ML models with **Python**, **FastAPI**, **AWS**, and **PostgreSQL**, serving **500** users and handling **10K** API requests per week
- Reduced dataset query latency by **70%** by designing a **PostgreSQL** metadata store with pointers to files in **AWS S3**, improving storage scalability and retrieval
- Reduced malformed datasets by **90%** by engineering an **LLM**-driven evaluator-optimizer workflow to score output quality and trigger retries with **NVIDIA NeMo Curator**, improving dataset quality and consistency
- Deployed the microservice on **AWS Elastic Beanstalk** with integrated load balancing and auto-scaling, achieving **99.9%** uptime and serving **10K** weekly API requests without manual infrastructure management

Software Engineering Intern

May 2025 – Aug 2025

Smart Cash Back TV Inc.

Toronto, ON

- Redesigned the **audio-fingerprinting** database using **C#** to reduce ad-recognition latency and improve reliability and scalability across the video analytics pipeline, allowing faster and more accurate audio matches
- Reduced query latency by **60%** and enabled **O(n)→O(1)** fingerprint-to-track resolution by introducing a reverse index and caching layer, eliminating repeated metadata scans

PERSONAL PROJECTS

SpecQA | *FastAPI, FAISS, PostgreSQL, AWS, GPT-4o*

- Building an agentic **RAG** pipeline with LLM-based evaluation for answer correctness, context relevance, and hallucination detection, aiming to dynamically optimize retrieval (top-k, re-ranking, prompt refinement) over a vectorized knowledge base of technical specifications and documentation

Syllabuddy | *Next.js, OpenAI API, Google Calendar API, Supabase, PyMuPDF*

syllabuddy.dev

- Built and launched a platform with **50** users that converts syllabus deadlines into exportable calendars
- Engineered an AI document processing pipeline to extract structured data from unstructured PDFs and images using **GPT-4o** with structured output parsing and automatic fallback to **OCR API** for image-based documents

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript, JavaScript, SQL, NoSQL, HTML, CSS

ML & Infrastructure: PyTorch, CUDA, Slurm, Docker, Kubernetes, Terraform, FFmpeg, NumPy, Pandas

Web & Backend: FastAPI, Node.js, Express, React.js, Next.js

Cloud, Data & Tools: AWS, GCP, Azure, PostgreSQL, MySQL, MongoDB, Redis, Git, GitHub Actions